

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12399529
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Lee; Seung Min et al.

Wearable device

Abstract

Provided is a wearable device. A wearable device according to an exemplary embodiment of the present invention comprises: a main body part including a display unit for augmented reality or virtual reality; a wearing part for maintaining the worn state of the main body part so that the display unit can be disposed on a user's face; a battery part coupled to the wearing part and including a first battery that serves as a main battery providing driving power to the main body part, and a second battery that serves as an auxiliary battery supporting the first battery; and an operation part provided in the wearing part to enable the removal of the second battery from the wearing part during a user's operation.

Inventors:	Lee; Seung Min (Gimpo-si, KR), Rho; Seung Yun (Gimpo-si, KR)
Applicant:	AMOGREENTECH CO., LTD. (Gimpo-si, KR)
Family ID:	1000008782606
Assignee:	AMOGREENTECH CO., LTD. (Gimpo-si, KR)
Appl. No.:	18/257978
Filed (or PCT Filed):	December 03, 2021
PCT No.:	PCT/KR2021/018193
PCT Pub. No.:	WO2022/131640
PCT Pub. Date:	June 23, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240053794 A1	Feb. 15, 2024

Foreign Application Priority Data

KR

10-2020-0177515

Dec. 17, 2020

Publication Classification

Int. Cl.: **G06F1/16** (20060101); **G06T19/00** (20110101); **H01M50/207** (20210101);
H01M50/247 (20210101); **H01M50/271** (20210101)

U.S. Cl.:

CPC **G06F1/163** (20130101); **G06T19/006** (20130101); **H01M50/207** (20210101);
H01M50/247 (20210101); **H01M50/271** (20210101); H01M2220/30 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9733480	12/2016	Baek et al.	N/A	N/A
10564433	12/2019	Lee	N/A	A63B 33/002
10580383	12/2019	Jiang	N/A	G02B 27/0176
10614745	12/2019	Lee	N/A	G09G 3/3266
10660224	12/2019	Wen	N/A	G02B 27/0176
10774987	12/2019	Xu	N/A	G06F 3/011
2015/0253574	12/2014	Thurber	359/630	G02B 27/0172
2016/0062125	12/2015	Baek	361/679.01	H05K 5/0086
2016/0357250	12/2015	Kim et al.	N/A	N/A
2020/0343775	12/2019	Yahagi	N/A	H02J 50/12

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2017229048	12/2016	JP	N/A
20160026557	12/2015	KR	N/A
1020160141982	12/2015	KR	N/A
20170136342	12/2016	KR	N/A
1020180109669	12/2017	KR	N/A
2017/123017	12/2016	WO	N/A

OTHER PUBLICATIONS

International Search Report issued in PCT/KR2021/018193 dated Mar. 17, 2022, 4 pgs. cited by applicant

Primary Examiner: Parker; Allen L

Assistant Examiner: Sasserath; Elisa

Attorney, Agent or Firm: ROTHWELL, FIGG, ERNST & MANBECK, P.C.

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application is a 35 U.S.C. 371 National Phase Entry Application from PCT/KR2021/018193 filed Dec. 3, 2021, which claims priority to and the benefit of Korean Patent Application No. 10-2020-0177515, filed on Dec. 17, 2020, the disclosures of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(2) The present invention relates to a wearable device.

BACKGROUND ART

(3) Recently, with the development of electronic technology, various means for displaying visual information to a user have been developed. As part of this, a display device that can be worn on a user's head is being developed.

(4) For example, a head mounted display (HMD) device is a display device capable of displaying visual information while worn on a user's head. Such an HMD device includes a virtual reality device that shows a virtual reality world through a display unit disposed at a position corresponding to the user's eyes when worn, an augmented reality device that shows an augmented screen together with a real world through the display unit, and the like.

(5) The HMD device requires a driving power to drive the display unit. To this end, the HMD device receives the driving power from a commercial power source through a power cable or from a battery built into the device itself.

(6) However, since the method of using the commercial power supplied through the power cable as the driving power restricts the user's activity, the method of embedding the battery in the device itself is used as the main method of providing the driving power.

(7) Accordingly, in order for the user to use the HMD device for a long time, a battery having a high storage capacity needs to be mounted on the HMD device. However, as the storage capacity of the battery increases, its size also increases. Therefore, since additional space is required to mount a larger capacity battery inside the HMD device, the overall size of the HMD device is also bound to increase.

(8) In addition, since the degree of freedom in design decreases when the space for mounting the battery increases, the diversity of design is inhibited.

DISCLOSURE

Technical Problem

(9) The present invention has been devised in view of the above points, and an object of the present invention is to provide a wearable device capable of implementing a reduction in weight and diversification of design even while increasing battery capacity.

Technical Solution

(10) In order to solve the above problems, the present invention provides a wearable device comprising: a body part including a display unit for augmented reality or virtual reality; a wearing part for maintaining a worn state of the body part so that the display unit can be disposed in front of a user's face; a battery part coupled to the wearing part and including a first battery serving as a main battery providing driving power to the body part and a second battery serving as an auxiliary battery supplementing the first battery; and an operation part provided in the wearing part to allow the second battery to be separated from the wearing part when operated by a user.

(11) In addition, the wearing part may include a frame formed in a ring shape to surround the user's head, and the frame may include: an arc-shaped first frame having one side connected to the body part and to which the first battery is coupled, an arc-shaped second frame in which both ends are respectively fastened to ends of the first frame and the operation part is provided, and an accommodation groove formed in the second frame to accommodate the second battery.

(12) In addition, the wearable device may further include a holder member in which the second battery is embedded, and the holder member may be detachably coupled to the accommodation groove.

(13) In addition, the wearable device may further include a cover member coupled to the first frame to protect the first battery disposed on one surface of the first frame from an external environment.

(14) Further, the body part may not include a battery for driving power.

(15) In addition, the wearing part may further include a plate-shaped support member coupled to the first frame, and the support member may be formed in a curved surface having a predetermined area to partially surround the user's head.

(16) In addition, each of the first battery and the second battery may be a flexible battery having flexibility.

(17) In this case, the flexible battery may include: an electrode assembly; an exterior material to encapsulate the electrode assembly together with an electrolyte; a first pattern formed in the electrode assembly such that mountain portions and valley portions are repeatedly arranged along the longitudinal direction of the electrode assembly; and a second pattern formed in the exterior material to coincide with the second pattern along the longitudinal direction of the exterior material.

Advantageous Effects

(18) According to the present invention, the overall weight can be reduced by using a thin flexible battery having a relatively light weight per storage capacity as a driving power source.

(19) In addition, according to the present invention, some of the batteries can be detachably coupled to the wearing part. Accordingly, when the power is completely exhausted in the wearable device according to an embodiment of the present invention, some of the batteries can be easily replaced, thereby increasing the usage time.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a view showing a wearable device according to an embodiment of the present invention.

(2) FIG. 2 is an exploded view of FIG. 1.

(3) FIG. 3 is a view showing a state in which the second battery is separated from FIG. 1.

(4) FIG. 4 is a view of FIG. 3 viewed from another direction.

(5) FIG. 5 is a schematic diagram showing a flexible battery that can be applied as a first battery and a second battery in a wearable device according to the present invention.

(6) FIG. 6 is an exemplary view showing various patterns applied to the electrode assembly and the exterior material in FIG. 5, in which valley portions or mountain portions adjacent to each other are shown at various intervals.

(7) FIG. 7 is an enlarged cross-sectional view showing a detailed configuration of FIG. 5.

MODES OF THE INVENTION

(8) Hereinafter, with reference to the accompanying drawings, embodiments of the present invention will be described in detail so as to be easily implemented by one of ordinary skill in the art to which the present invention pertains. The present invention may be embodied in a variety of forms and is not be limited to the embodiments described herein. In order to clearly describe the present invention in the drawing, parts irrelevant to the description are omitted from the drawings; and throughout the specification, same or similar components are referred to as like reference numerals.

(9) An wearable device **100** according to an embodiment of the present invention may be a head mounted display (HMD) device for implementing a visual virtual reality or augmented reality.

(10) When the wearable device **100** is worn on a user's head, a display unit **112** may be disposed at a position corresponding to the user's eyes, and the user may check various information or images displayed through the display unit **112**.

(11) To this end, as shown in FIGS. 1 to 4, the wearable device **100** according to an embodiment of the present invention may include a body part **110**, a wearing part **120**, a battery part **130a** and **130b**, and an operation part **140**.

(12) When the user wears the wearable device **100** on the head through the wearing part **120**, the body part **110** may be disposed at a position corresponding to the user's eyes.

(13) Such a body part **110** may provide various image information to the user.

(14) To this end, as shown in FIG. 1, the body part **110** may include a display unit **112** for displaying various image information, and a control part for controlling an operation of the display

unit **112**.

(15) Herein, the control part may be a microprocessor, and the display unit **112** may be an augmented reality display on which augmented reality is output, or a virtual reality display on which virtual reality is output.

(16) Further, the display unit **112** may be a transparent organic light emitting display.

(17) Accordingly, the user may experience through the display unit **112** the augmented reality in which a real world viewed through the transparent organic light emitting display is overlapped with information or images output through the transparent organic light emitting display, or the virtual reality that imitates a specific environment or situation in reality as it is and makes the user seem to interact with the real surrounding situation or environment.

(18) Meanwhile, the body part **110** may further include other functional modules for performing various functions in addition to the display unit **112**. For example, the body part **110** may further include at least one of: a communication module for wirelessly transmitting and receiving data, various sensor modules for detecting the user's gestures, movements, temperature, speed, etc., a camera module for obtaining image information around the user, a microphone module and a speaker module.

(19) In addition, the body part **110** may further include a data storage unit for storing information obtained through the above-described functional modules, and a USB port for connection with an external storage medium.

(20) As described above, when the user wears the wearable device **100** according to an embodiment of the present invention, the wearing part **120** may fix the body part **110** so that it is located in front of the user's face.

(21) For example, the wearing part **120** may be provided as a frame formed in a ring shape to surround the user's head when the wearable device **100** is worn.

(22) As a non-limiting example, the frame may include an arc-shaped first frame **121** having one side connected to the body part **110**, and an arc-shaped second frame having both ends fastened to ends of the first frame **121**, respectively, and the first frame **121** and the second frame **122** may be disposed to be convex in opposite directions.

(23) Accordingly, the frame may be formed in a ring shape by combining the first frame **121** and the second frame **122**, and when the wearable device is worn, one surface of each of the first frame **121** and the second frame **122** facing each other may surround the user's head while being in contact with the user's head.

(24) Accordingly, when the user wears the wearable device **100** according to an embodiment of the present invention, the wearing part **120** may surround the user's head, and the body part **110** connected to one side of the first frame **121** may be disposed to locate in front of the user's face.

(25) Herein, both ends of the second frame **122** may be fixedly coupled to the ends of the first frame **121**, but may be rotatably coupled to the first frame **121**.

(26) In this case, the wearable device **100** according to an embodiment of the present invention may be prevented from flowing downward and releasing the worn state while the wearing part **120** is disposed to surround the user's head.

(27) To this end, the wearing part **120** may further include a plate-shaped support member **125** coupled to the first frame **121**. Such a support member **125** may be provided to have a predetermined area to partially surround the user's head, and may be formed in a curved surface so as to be in close contact with the user's head.

(28) Accordingly, when the user wears the wearable device **100** according to an embodiment of the present invention, the first frame **121** and the second frame **122** may surround the user's head, and the support member **125** may be in contact with the upper portion of the user's head. Therefore, the support member **125** may prevent the first frame **121** and the second frame **122** from flowing downward while being worn on the user's head.

(29) The battery part **130a** and **130b** may serve as a power supply source for supplying driving

power to the body part **110**.

(30) That is, the battery part **130a** and **130b** may provide the driving power to a control part (not shown) built in the body part **110**, so that various functional modules including the display unit **112** contained in the body part **110** may be driven under the control of the control part.

(31) Such a battery part **130a** and **130b** may be embedded in the wearing part **120** to prevent the weight from being concentrated on the body part **110** disposed in front of the user's face when the wearable device **100** according to an embodiment of the present invention is worn, and may be configured as a flexible battery so as to have high capacity while reducing the overall weight.

(32) That is, in the wearable device **100** according to an embodiment of the present invention, the battery part **130a** and **130b** serving as a power supply source is not built into the body part **110**, but may be embedded in the wearing part **120** surrounding the user's head so that the worn state of the body part **110** may be maintained.

(33) For example, the battery part **130a** and **130b** may be provided with at least one flexible battery and be coupled to the wearing part **120** so as to be disposed along the longitudinal direction of the first frame **121** and/or the second frame **122** constituting the wearing part **120**.

(34) Accordingly, in the wearable device **100** according to an embodiment of the present invention, the weight of the body part **110** itself may be reduced by the weight of the omitted battery, whereby the user's body may be prevented from being tilted forward by the weight of the body part **110** even when the user wears the wearable device **100**. Further, the total weight of the wearable device **100** may be shared because the weight of the wearing part **120** itself may be increased by the weight of the added battery.

(35) In addition, in the wearable device **100** according to an embodiment of the present invention, since the battery part **130a** and **130b** is disposed in the wearing part **120**, the body part **110** may not require a separate space for embedding a battery. Accordingly, since the total size of the body part **110** may be reduced, the wearable device **100** according to an embodiment of the present invention may be implemented with a miniaturization or thinness.

(36) In addition, since the mounting space of the battery is omitted in the body part **110**, the internal space of the body part **110** may be usefully utilized if the size of the body part is kept the same as in the prior art. Accordingly, other additional functional modules in addition to the aforementioned functional modules may be further included in the internal space of the body part **110**.

(37) In addition, the wearable device **100** according to an embodiment of the present invention may use, as a power supply source, a flexible battery having a relatively light weight per storage capacity instead of a prismatic battery having a relatively heavy weight per storage capacity, thereby reducing the overall weight of the device and implementing a reduction in weight.

(38) In this case, the battery part **130a** and **130b** may be provided with a single flexible battery, but may be provided with a plurality of flexible batteries to increase the total capacity. In addition, at least one flexible battery may be detachably coupled to the wearing part **120** so that the battery part **130a** and **130b** may be continuously used through simple replacement even when the battery power is exhausted.

(39) For example, the battery part **130a** and **130b** may include a first battery **130a** serving as a main battery providing driving power to the body part **110** and a second battery **130b** serving as an auxiliary battery supplementing the first battery **130a**, and the second battery **130b** may be detachably coupled to the wearing part **120**, and the second battery **130b** may be easily separated from the wearing part **120** by a user's manipulation.

(40) Herein, the first battery **130a** and the second battery **130b** may be provided as a flexible battery as described above, and the flexible battery may be a plate-shaped flexible battery having a predetermined area and length. In addition, each of the first battery **130a** and the second battery **130b** may further include separate circuit units **152** and **153** for controlling charging/discharging or preventing overcharging. A detailed description of such a flexible battery will be provided later.

(41) Accordingly, when the power of the battery part **130a** and **130b** is partially or completely exhausted, the user may replace the second battery **130b** through a simple manipulation to provide driving power to the first battery **130a**. Accordingly, the body part **110** may be continuously driven.

(42) To this end, the wearing part **120** may include an accommodation groove **123** for accommodating the second battery **130b**, and the second battery **130b** may be detachably coupled to the accommodation groove **123**. In addition, the wearable device **100** may include a operation part **140** provided in the wearing part **120** so that the second battery **130b** mounted in the accommodation groove **123** may be easily separated from the accommodation groove **123** by a user's manipulation.

(43) As a non-limiting example, as shown in FIG. 2, the second frame **122** may include an accommodation groove **123** formed along the longitudinal direction, and the operation part **140** may be provided in the second frame **122** so as to be positioned around an edge defining the accommodation groove **123**.

(44) In addition, the accommodation groove **123** may be formed to be recessed inwardly from an upper surface of the second frame **122** by a predetermined depth so that the upper portion thereof is opened, and the operation part **140** may be provided as a button capable of being pressed by a user on one surface of the second frame **122**.

(45) In this case, as shown in FIGS. 1 and 2, the first battery **130a** may be disposed on one surface of the first frame **121** along the longitudinal direction of the first frame **121**, and the second battery **130b** may be inserted into the accommodation groove **123**.

(46) In this case, the first battery **130a** may be disposed on one surface of the first frame **121** via a cover member **124** coupled to one surface of the first frame **121**, and may be electrically connected to the body part **110**.

(47) In addition, the second battery **130b** may be detachably coupled to the accommodation groove **123** in a state of being coupled to a holder member **151**.

(48) Herein, the second battery **130b** may be formed integrally with the holder member **151** or may be detachably coupled to one side of the holder member **151**.

(49) In addition, the holder member **151** may be formed of a material having rigidity so that a shape substantially similar to that of the accommodation groove **123** may be maintained, but may also be formed of a soft material having flexibility.

(50) In addition, the holder member **151** may be electrically connected to the first battery **130a** while being inserted into the accommodation groove **123**. To this end, the holder member **151** may include a connection terminal exposed to the outside so as to be electrically connected to the second battery **130b**, and the connection terminal may be in contact with a corresponding connection terminal provided inside the accommodation groove **123**. For example, any one of the connection terminal and the corresponding connection terminal may be a known pogo pin.

(51) Accordingly, the first battery **130a** may be prevented from being exposed to the outside by the cover member **124**, and may serve as a main battery for providing driving power to the body part **110** while being fixed to one side of the first frame **121**.

(52) In addition, the second battery **130b** may be electrically connected to the first battery **130a** to provide power thereto while being mounted in the accommodation groove **123** via the holder member **151**. Accordingly, the second battery **130b** may serve as an auxiliary battery that supports the capacity of the first battery **130a**, and may be separated from the accommodation groove **123** and replaced with a new second battery **130b** when some or all of the stored power is exhausted.

(53) That is, as shown in FIGS. 3 and 4, when the user operates the operation part **140** in case the second battery **130b** mounted in the accommodation groove **123** needs to be replaced, the second battery **130b** may move upward from the accommodation groove **123** in a state of being coupled to the holder member **151** so that at least a portion of the length thereof may protrude to the outside.

(54) Accordingly, in the wearable device **100** according to an embodiment of the present invention, the second battery **130b** detachably coupled to the accommodation groove **123** may function as an

auxiliary battery that supplies power to the first battery **130a**. Therefore, in the wearable device **100** according to an embodiment of the present invention, power may be continuously supplied to the first battery **130a** through the replacement of the second battery **130b** even if the power of the first battery **130a** and the second battery **130b** is completely exhausted.

(55) As a result, the wearable device **100** according to an embodiment of the present invention may be continuously driven by replacing the second battery **130b**, thereby increasing the use time or enabling continuous use.

(56) Herein, the second battery **130b** may supply the stored power to the first battery **130a**, but may also directly supply the stored power to the body part **110** when the power of the first battery **130a** is completely exhausted.

(57) Meanwhile, the first battery **130a** and the second battery **130b** may be provided as a flexible battery having flexibility as described above, and the flexible battery may be a flexible plate shape having a predetermined area and length.

(58) Accordingly, even if the first battery **130a** and the second battery **130b** are coupled to the first frame **121** and the second frame **122** formed in an arc shape, respectively, the first battery **130a** and the second battery **130b** may be changed to correspond to the shape of the first frame **121** and the second frame **122**.

(59) Specifically, the first battery **130a** and the second battery **130b** may be respectively coupled to the first frame **121** and the second frame **122** in a state of being bent in an arc-shape as shown in FIG. 2.

(60) The first battery **130a** and the second battery **130b** may be in a form in which an electrode assembly **131** is encapsulated together with an electrolyte inside exterior materials **137** and **138** as shown in FIGS. 5 to 7.

(61) In this case, each of the first battery **130a** and the second battery **130b** may include patterns **136** and **139** formed along the longitudinal direction of the electrode assembly **131** and the exterior materials **137** and **138**, and the first pattern **136** formed in the electrode assembly **131** and the second pattern **139** formed in the exterior materials **137** and **138** may be provided to have the same directionality.

(62) Accordingly, as shown in FIGS. 1 to 4, even though the first battery **130a** and the second battery **130b** are mounted in the wearing part **120** in a bent state, an amount of deformation of the base material itself constituting the electrode assembly **131** and the exterior materials **137** and **138** in the first battery **130a** and the second battery **130b** may be prevented or minimized through the patterns **136** and **139**.

(63) As such, even if the first battery **130a** and the second battery **130b** are mounted in a bent state, the amount of deformation of the base material itself that may occur at the bent portion can be minimized, whereby the electrode assembly **131** and the exterior materials **137** and **138** in the first battery **130a** and the second battery **130b** can be prevented from being damaged or deteriorated due to bending.

(64) In this case, the first pattern **136** and the second pattern **139** may be respectively formed in the electrode assembly **131** and the exterior materials **137** and **138** so as to coincide with each other as well as to have the same directionality. This is to allow the first pattern **136** and the second pattern **139** to always have the same behavior.

(65) As described above, in the first battery **130a** and the second battery **130b** applied to the present invention, the patterns **136** and **139** respectively formed in the electrode assembly **131** and the exterior materials **137** and **138** for the contraction and relaxation in the longitudinal direction are formed to coincide with each other, whereby the electrode assembly **131** and the exterior materials **137** and **138** may always maintain a uniform interval or contact state over the entire length even though bending in the longitudinal direction occurs.

(66) Accordingly, even if the first battery **130a** and the second battery **130b** are respectively mounted in the first frame **121** and the second frame **122** in a bent state, the electrolyte packed

together with the electrode assembly **131** may be uniformly distributed over the entire length, thereby preventing the performance as a battery from being deteriorated.

(67) For example, as shown in FIG. **6(a)** to FIG. **6(d)**, each of the first pattern **136** and the second pattern **139** may have mountain portions and valley portions formed in a direction parallel to the width direction of the electrode assembly **131** and the exterior materials **137** and **138**, and the mountain portions and valley portions may be alternately disposed along the longitudinal direction of the electrode assembly **131** and the exterior materials **137** and **138**.

(68) In addition, the mountain portions and valley portions constituting the first pattern **136** and the second pattern **139** may be formed at the same positions as the mountain portions and valley portions, respectively. Accordingly, the first pattern **136** and the second pattern **139** may coincide with each other.

(69) Specifically, the mountain portions and valley portions of the first pattern **136** and the second pattern **139** may be formed in a direction parallel to a straight line parallel to the width direction of the electrode assembly **131** and the exterior materials **137** and **138**, and may be formed to be repeatedly disposed along the longitudinal direction.

(70) In this case, the patterns **136** and **139** may be continuously or discontinuously formed in a direction parallel to the width direction of the electrode assembly **131** and the exterior materials **137** and **138**. In addition, the patterns **136** and **139** may be formed over the entire length of the electrode assembly **131** and the exterior materials **137** and **138**, or may be partially formed over a partial length.

(71) Further, the mountain portions and valley portions may be provided to have various cross sections, such as an arc-shaped cross section including a semicircle, a polygonal cross section including a triangle or a quadrangle, and a cross section in which the arc-shaped cross section and the polygonal cross section are combined. In addition, each of the mountain portions and valley portions may be provided to have the same pitch and width, but may also be provided to have different pitches and widths.

(72) Accordingly, the first battery **130a** and the second battery **130b** may reduce fatigue applied to the base material itself through the patterns **136** and **139** even though the electrode assembly **131** and the exterior materials **137** and **138** are embedded in the wearing part **120** in a bent state, thereby preventing the battery from being damaged or deteriorating in performance as a battery.

(73) Meanwhile, the first pattern **136** and the second pattern **139** may be provided so that intervals between adjacent mountain portions or between adjacent valley portions are formed at the same distance, different distances, or a combination thereof.

(74) The electrode assembly **131** may be encapsulated together with the electrolyte inside the exterior materials **137** and **138**. As shown in FIG. **7**, the electrode assembly **131** may include a positive electrode **132**, a negative electrode **134**, and a separator **133**.

(75) The positive electrode **132** may include a positive electrode current collector **132a** and a positive electrode active material **132b**, the negative electrode **134** may include a negative electrode current collector **134a** and a negative electrode active material **134b**, and each of the positive electrode current collector **132a** and the negative electrode current collector **134a** may be implemented in the form of a plate-shaped sheet having a predetermined area.

(76) That is, the positive electrode **132** and the negative electrode **134** may have active materials **132b** and **134b** compressed, deposited or applied on one or both surfaces of the current collectors **132a** and **134a**, respectively. In this case, the active materials **132b** and **134b** may be provided over the entire area of the current collectors **132a** and **134a**, respectively, or may be provided partially over a partial area thereof.

(77) Further, the positive electrode current collector **132a** and the negative electrode current collector **134a** may include a negative electrode terminal **135a** and a positive electrode terminal **135b** for electrical connection to an external device, respectively.

(78) Herein, the positive electrode terminal **135b** and the negative electrode terminal **135a** may be

provided to extend from the positive electrode current collector **132a** and the negative electrode current collector **134a**, respectively, by a predetermined length to protrude from one sides of the exterior materials **137** and **138**, or to be exposed on the surfaces of the exterior materials **137** and **138**.

(79) In this case, the positive electrode active material **132b** and the negative electrode active material **134b** may include a polytetrafluoroethylene (PTFE) ingredient. Accordingly, even if the first battery **130a** and the second battery **130b** are bent, the positive electrode active material **132b** and the negative electrode active material **134b** may be prevented from peeling off from the current collectors **132a** and **134a**, respectively, or from cracking due to the bending.

(80) Meanwhile, the separator **133** disposed between the positive electrode **132** and the negative electrode **134** may include a nanofiber web layer **151b** disposed on one or both surfaces of a non-woven fabric layer **151a**.

(81) Herein, the nanofiber web layer **151b** may be a nanofiber containing at least one selected from a polyacrylonitrile nanofiber and a polyvinylidene fluoride nanofiber.

(82) The exterior materials **137** and **138** may be formed of a plate-shaped member having a predetermined area, and may protect the electrode assembly **131** from an external force by accommodating the electrode assembly **131** and the electrolyte therein.

(83) As an example, the exterior materials **137** and **138** may include a pair of first exterior material **137** and second exterior material **138**, and the pair of first exterior materials **137** and second exterior materials **138** may be sealed with an adhesive on the edges abutting each other. Accordingly, the electrolyte and electrode assembly **131** accommodated inside the exterior materials **137** and **138** may be prevented from being exposed to the outside, and the electrolyte may be prevented from leaking to the outside.

(84) In the exterior materials **137** and **138**, the first exterior material **137** and the second exterior material **138** may be formed of two members, and all edges of the first exterior material **137** and the second exterior material **138** constituting a sealing portion may be sealed with an adhesive. Alternatively, the first exterior material **137** and the second exterior material **138** may be formed of one member and folded in half along a width direction or a longitudinal direction thereof, and the remaining edge portions of the first exterior material **137** and the second exterior material **138** abutting each other may be sealed with an adhesive.

(85) Meanwhile, the wearable devices **100** according to the present invention may be implemented in a form that serves as both an augmented reality function and a virtual reality function, may be a form in which the augmented reality function and the virtual reality function are integrated, or may display information linked with a smartphone on the display unit **112**.

(86) In addition, when the wearable device **100** according to the present invention is implemented as a virtual reality device, a smartphone having a virtual reality function may be detachably coupled to the body part **110**. In this case, the display unit **112** included in the body part **110** may be omitted, and the smartphone coupled to the body part **110** may replace the display unit **112**.

(87) Although an embodiment of the present invention have been described above, the spirit of the present invention is not limited to the embodiment presented in the subject specification; and those skilled in the art who understands the spirit of the present invention will be able to easily suggest other embodiments through addition, changes, elimination, and the like of elements without departing from the scope of the same spirit, and such other embodiments will also fall within the scope of the present invention.

Claims

1. A wearable device comprising: a body part including a display unit for augmented reality or virtual reality; a wearing part for maintaining a worn state of the body part so that the display unit can be disposed in front of a user's face; a battery part coupled to the wearing part, and including a

first battery serving as a main battery providing driving power to the body part and a second battery serving as an auxiliary battery supplementing the first battery; and an operation part provided in the wearing part to allow the second battery to be separated from the wearing part when operated by the user, wherein the wearing part comprises a frame having a ring shape to surround the user's head, wherein the frame comprises: an arc-shaped first frame having one side connected to the body part, and to which the first battery is coupled; and an arc-shaped second frame, each end of which is respectively fastened to a corresponding end of the arc-shaped first frame, wherein the operation part is disposed in the arc-shaped second frame, and wherein the arc-shaped second frame comprises an accommodation groove configured to accommodate the second battery.

2. The wearable device of claim 1, wherein the wearable device further includes a holder member in which the second battery is embedded, and wherein the holder member is detachably coupled to the accommodation groove.

3. The wearable device of claim 1, wherein the wearable device further includes a cover member coupled to the arc-shaped first frame to protect the first battery disposed on one surface of the arc-shaped first frame from an external environment.

4. The wearable device of claim 1, wherein the body part does not include a battery for driving power.

5. The wearable device of claim 1, wherein the wearing part further includes a plate-shaped support member coupled to the arc-shaped first frame, and wherein the support member is formed in a curved surface having a predetermined area to partially surround the user's head.

6. The wearable device of claim 1, wherein each of the first battery and the second battery is a flexible battery having flexibility.

7. The wearable device according to claim 6, wherein the flexible battery includes: an electrode assembly; an exterior material to encapsulate the electrode assembly together with an electrolyte; a first pattern formed in the electrode assembly such that mountain portions and valley portions are repeatedly arranged along the longitudinal direction of the electrode assembly; and a second pattern formed in the exterior material to coincide with the first pattern along the longitudinal direction of the exterior material.
